

67 frequently asked questions across 13 categories, with scientifically accurate student-friendly answers

FAQ compendium

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All datasets used in this package are simulated teaching data. Wet-lab and animal procedures described here are conceptual; institutional biosafety approval and SOPs govern all practice.

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FAQs — GMOs, Cloning & Gene Expression

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A. Beginner (1–6)

1. What is a GMO in one sentence?

An organism whose genetic material has been altered by humans using laboratory techniques — though whether *any particular* altered organism is legally a "GMO" depends on the country.

2. Is genetic engineering the same as breeding?

No. Breeding recombines existing variation through controlled mating; genetic engineering introduces designed changes directly, often across species boundaries or at single-base precision.

3. Are GM foods "chemicals"?

All food is chemicals. The meaningful questions are which trait was changed and what the safety assessment showed — not the word "chemical".

4. Why do bacteria have restriction enzymes if we use them for cloning?

They are a bacterial defense against phage DNA; bacteria protect their own DNA with methylation. We borrowed the enzymes as molecular scissors.

5. What's the difference between a plasmid and a vector?

A plasmid is a naturally occurring circular DNA replicon; a *vector* is a plasmid (or virus/artificial chromosome) engineered to carry DNA we care about.

6. Do GMOs always contain foreign DNA?

No — gene-edited organisms may carry only a small deletion or base change, with no foreign sequence remaining.

B. Molecular cloning (7–15)

7. Why is an origin of replication essential?

Without an *ori*, the plasmid can't be copied by the host and is lost — no propagation, no colonies.

8. Why do we need both a promoter and a terminator?

The promoter starts transcription; the terminator stops it at a defined point, preventing read-through into vector sequences that can destabilize the construct.

9. What does "high-copy plasmid" mean practically?

The *ori* (e.g., pUC-derived) allows hundreds of plasmid copies per cell → more DNA per prep, but more metabolic burden — problematic for toxic or unstable inserts.

10. Why add extra bases (clamps) before restriction sites in primers?

Enzymes need flanking DNA to bind and cut efficiently; sites at the very end of a PCR product are cut poorly.

11. Can I ligate an EcoRI end to a BamHI end?

No — overhangs must be complementary (AATT vs GATC don't pair). Compatible but different enzymes (e.g., BamHI/BglII) *can* ligate because they leave identical overhangs.

12. Why dephosphorylate the vector?

Removes 5' phosphates so the vector can't self-ligate; only insert-containing molecules (providing the phosphates) transform efficiently.

13. What is an MCS?

Multiple cloning site — a synthetic cluster of unique restriction sites so many different enzyme pairs can target the same small region.

14. What's a shuttle vector?

A vector with two origins and two selectable markers so it propagates in two hosts (e.g., build in *E. coli*, use in yeast or mammalian cells).

15. What does "in-frame" mean for a fusion?

The insert joins the tag coding sequence without a frameshift, so one continuous reading frame produces a single fusion protein.

C. Cloning troubleshooting (16–24)**16. No colonies at all — first suspects?**

Dead competent cells, wrong antibiotic, failed ligation/digest. Always run a positive-control transformation.

17. All colonies are empty vector — why?

Vector self-ligation (missing dephosphorylation), incompatible insert ends, or insert preparation failed. Check the vector-only control plate count.

18. Low ligation efficiency — what to tune?

Molar ratio (~3:1), ligase freshness, buffer ATP, compatible ends, insert purity (salts inhibit ligase).

19. My PCR product won't clone by TA — why?

High-fidelity polymerases leave blunt ends; you must A-tail the product (or use blunt-end/Gibson approaches).

20. Gibson assembly gave many colonies but wrong plasmids — likely cause?

Repeats in overlaps causing misassembly, or overlaps too short/weak; re-design overlaps with 20–40 bp unique sequence.

21. Golden Gate didn't work — check what first?

Internal Type IIS sites in your parts (undomesticated), wrong fusion-site map, or enzyme temperature cycling not followed.

22. Colony PCR gives smeared bands — why?

Too much colony template (inhibitors), dirty cells, or primer issues; use a tiny amount of colony and fresh mix.

23. Digest shows three bands but map predicts two — panic?

First sum the fragments; if sum > plasmid size → rearrangement. If sum matches → an extra site (methylation? star activity? re-check sequence).

24. Sanger trace is mixed/overlapping — what does it mean?

Heterozygous site (rare in plasmids), mixed colony, or indel mixture from a repair event; re-isolate a single colony and re-prep.

D. GMO (25–31)

25. What's the first GMO ever made?

Functionally cloned recombinant plasmids in *E. coli* — Cohen, Chang, Boyer & Helling, 1973 (the modern GMO era's starting point).

26. Is Golden Rice transgenic?

Yes — it carries *psy* and *crtI* genes from other species (maize/daffodil/*Erwinia*), inserted by transformation.

27. Why do Bt crops need refuges?

Refuges maintain susceptible insects that mate with any resistant survivors, slowing resistance evolution — resistance management, not an admission of failure.

28. Can a GMO be made without inserting anything?

Yes — a knockout via CRISPR NHEJ deletes bases; a base edit changes a letter. No foreign DNA remains.

29. What's the difference between transfection, transformation, and transduction?

Transfection = introducing DNA into eukaryotic cells; transformation = into bacteria (usage differs by field); transduction = viral-mediated delivery.

30. Are all transgenic animals made by microinjection now?

No — CRISPR reagent delivery to embryos is now common for edits; transgenesis (added DNA) still uses microinjection, transposons, viral vectors, or SCNT depending on species and goal.

31. Why is plant transformation "easier" conceptually?

Totipotency — one modified somatic cell can regenerate a whole plant via tissue culture; animal embryos must be manipulated individually.

E. CRISPR (32–40)

32. What is a PAM and why does it matter?

Protospacer-adjacent motif — a short sequence (e.g., NGG for SpCas9) next to the target that Cas9 must recognize to initiate binding; it's required for cleavage and limits targetable sites.

33. Does CRISPR always cut?

Not necessarily — dCas9 variants bind without cutting, powering CRISPRa/i; base and prime editors avoid DSBs by design.

34. NHEJ vs HDR in one line each?

NHEJ: error-prone religation of broken ends → indels (knockout). HDR: template-guided repair → precise insertion/correction (needs donor, cell-cycle dependent).

35. Why are knock-ins hard?

HDR is inefficient in most primary cells; delivery of donors is limiting; alternatives (HITI, base/prime editing) have their own constraints.

36. What is mosaicism?

Different cells of the same founder animal carry different edits because editing happened after the first division — complicates genotyping and phenotyping.

37. Are off-targets a deal-breaker?

They're manageable: careful guide design, high-fidelity variants, transient RNP delivery, and empirical verification of predicted off-target sites.

38. CRISPRi vs CRISPRa?

CRISPRi (dCas9-KRAB) represses transcription — tests *necessity*; CRISPRa (dCas9-activators) up-regulates — tests *sufficiency*, all without genome change.

39. Can CRISPR edit RNA?

Cas13-class systems target RNA without DNA changes — used for knockdown and RNA-editing research tools.

40. Why did the 2018 human-embryo case draw global condemnation?

Germline editing of humans without adequate oversight/consent framework, questionable benefit-risk, and transparency failures — condemned irrespective of the technical outcome.

F. Reporter genes (41–46)

41. Why does GFP need no substrate?

Its chromophore forms autocatalytically from internal amino acids (needs only O₂ to mature) — that's why it revolutionized live imaging.

42. When would I choose luciferase over GFP?

When you need *quantitative* activity over time (plate-reader assays, dual-reporter normalization) rather than spatial detail.

43. Why does my GFP fusion not localize like the endogenous protein?

The tag can interfere (steric effects, missing targeting signals, oligomerization of old FPs) — validate against IF/ISH and consider tag placement/linkers.

44. What is a destabilized GFP?

GFP fused to a degradation tag so it degrades quickly — reports *current* transcription rather than accumulated history.

45. Why normalize luciferase to Renilla?

Transfection efficiency and cell number vary between wells; Renilla from a co-transfected constitutive plasmid calibrates that variation.

46. Can reporters mislead about endogenous expression?

Yes — missing chromatin context or distal elements; always pair with endogenous methods (ISH/IF) for strong claims.

G. Developmental biology (47–52)

47. What is a morphogen in one line?

A signaling molecule whose concentration gradient instructs different cell fates at different thresholds (Wolpert's French-flag model).

48. Why are Hox genes "colinear"?

Their order on the chromosome matches the order (anterior→posterior) and timing of their expression along the body axis.

49. What did the *myoD* experiment show?

Forced *myoD* expression converts fibroblasts into myoblasts — a landmark *sufficiency* demonstration (with the caveat that endogenous specification is multi-input).

50. What is lateral inhibition?

Notch-mediated neighbor interactions where one cell's fate suppresses the same fate in adjacent cells — producing salt-and-pepper patterns (e.g., neural precursors).

51. Why is the zebrafish embryo so useful?

External, transparent, rapid development → live imaging of every stage; large clutches; forward screens.

52. What's a "master regulator" — and the caveat?

A gene whose expression is sufficient to induce a whole program (e.g., *myoD*); caveat — endogenous development usually requires multiple inputs, so "master" describes the experimental sufficiency, not the whole mechanism.

H. Spatial gene expression (53–57)**53. Spatial vs temporal expression — difference?**

Spatial = where in the organism/tissue; temporal = when in development/response; spatiotemporal = both jointly.

54. Gradient vs domain?

A gradient is a continuous concentration change across space; a domain is a sharply bounded region of expression. Gradients inform; domains execute.

55. Why does bulk RNA-seq lose spatial information?

It averages transcripts over the whole sampled tissue — a strong signal can come from a tiny, highly expressing region.

56. What does scRNA-seq lose and why?

Position — cells are dissociated into a suspension before barcoding, severing each cell from its coordinates (motivating spatial transcriptomics).

57. What is spot deconvolution?

Array-based spatial data measures mixtures of cells per spot; deconvolution estimates the underlying cell-type proportions using single-cell references.

I. Experimental methods (58–61)**58. Why is the sense probe the right negative control for ISH?**

It has the same sequence as the target mRNA and cannot base-pair specifically — it measures everything except intended hybridization.

59. RT-PCR vs RT-qPCR?

Both start with reverse transcription; RT-PCR is endpoint (present/absent, semi-quantitative); RT-qPCR monitors amplification in real time for quantitative comparison.

60. Why validate housekeeping/reference genes per experiment?

"Housekeeping" genes can shift with treatment, stage, or tissue — an unvalidated reference invalidates the $\Delta\Delta C_t$ result.

61. IHC vs ISH — when each?

IHC detects the *protein* (post-transcriptional regulation visible); ISH detects the *mRNA* (transcriptional pattern); they disagree meaningfully when translation/stability is regulated.

J. Data analysis (62–63)

62. Why do statistics on ΔC_t rather than fold change?

C_t values are approximately normally distributed; fold changes are skewed — analyzing ΔC_t keeps the statistics valid.

63. What's FDR and why not just p-values?

With thousands of genes tested, 5% of "significant" hits are expected by chance; Benjamini–Hochberg FDR controls the *expected proportion of false discoveries* among your calls.

K. Ethics (64)

64. Are GMOs "unnatural" in a scientifically meaningful sense?

"Natural" is not a scientific property — the scientific questions are what change was made, what the evidence says, and what the risks are; "unnaturalness" belongs to the ethics/values layer (Module 14), and both risk and benefit arguments must be made explicitly, not by label.

L. Advanced concepts (part of 65)

65. Quick definitions: TAD? Super-enhancer? Bursting? Pseudotime?

- **TAD:** self-interacting chromatin domain constraining enhancer–promoter contacts.
- **Super-enhancer:** dense enhancer cluster with high Mediator/BRD4/H3K27ac controlling identity genes.
- **Bursting:** intermittent transcription initiation producing cell-to-cell mRNA-number variability.
- **Pseudotime:** ordering single cells along a trajectory of transcriptional similarity — developmental sequence, not clock time.

M. Exam preparation

66. How should I answer "compare X and Y" questions?

Table form: definition → mechanism → strength → limitation → example, then a one-sentence verdict tied to the scenario in the question.

67. How do I avoid overclaiming in experimental answers?

Name the readout's resolution (space/time/quantity), the controls that were or weren't run, and what a *stronger* claim would require (validation method).

(Total: 67 Q&As — exceeds the 60-question requirement.)